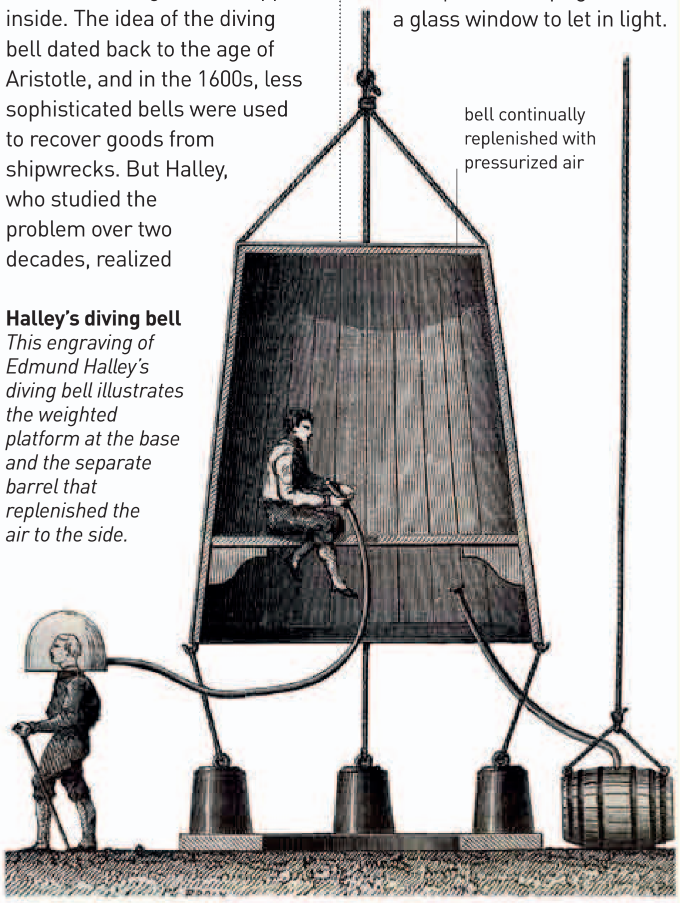
him, but we now know the disease is caused by a parasite spread by female *Anopheles* mosquitoes (see 1893–94).

In England, astronomer **Edmond Halley** made the first safe and practical **diving bell**—a bell-shaped diving chamber that enabled a person to go under water, breathing the air trapped inside. The idea of the diving bell dated back to the age of Aristotle, and in the 1600s, less sophisticated bells were used to recover goods from shipwrecks. But Halley, who studied the problem over two decades, realized that air is compressed by water pressure at depth, which is why a simple air tube to the surface did not work. Halley's ingenious solution was to continually replenish the air in the bell with air pressurized in weighted barrels lowered beside the bell. He also added a weighted tray to keep the bell upright, and a glass window to let in light.



Halley's diving bell
This engraving of Edmond Halley's diving bell illustrates the weighted platform at the base and the separate barrel that replenished the air to the side.

1717–20

The 18th century saw the creation of the first scientific collections of butterflies, like this collection of British butterflies, first named by English naturalist James Petiver.

SMALLPOX WAS A DEADLY DISEASE IN THE 18TH CENTURY. Millions of people, many of them children, died from the illness, and even those who recovered were left with faces permanently disfigured by the scars. Yet long ago, the Chinese had noticed that once people had survived smallpox, they never caught it again, no matter how much they were exposed to the disease. Chinese physicians began deliberately rubbing infected material into a scratch on healthy people. Some died quickly from the infection this caused, but most survived, and seemed to gain immunity to the disease. The **practice of "variolation,"** as it became known, spread across Asia to Turkey, where it was noticed by Greek physician **Giacomo Pylari** (1659–1718), and then the young wife of the British ambassador to Constantinople (Istanbul), **Lady Mary Wortley Montagu** (1689–1762). Montagu was so impressed that she wrote a famous series of letters home advocating its use. She had her own children inoculated, and campaigned ardently to introduce the practice to the British upper classes. Her pioneering efforts led Edward Jenner to discover vaccination (see 1796).

LADY MARY MONTAGU
(1689–1762)

Mary Montagu's campaign to introduce smallpox inoculation in Britain helped to establish the idea that disease could be prevented through immunity. She had had smallpox herself as a young woman. Besides her pioneering work on disease, she was a celebrated writer, much admired by some of the leading figures of the day.

In 1717, London apothecary **James Petiver** (1685–1718) published *Papilionum Brittaniae Icones* (*Images of British Butterflies*). It was one of the **first great catalogs of butterflies**, based on Petiver's collection of species, now in London's Natural History Museum.

In 1718, English inventor James Puckle (1667–1724) was working on the design of a forerunner of the machine gun. The **Puckle gun** was a flintlock rifle mounted on a tripod with a revolving cylinder holding 11 shots that could be turned by a handle to fire 63 shots in 7 minutes—three times as fast as the best musketman.

Butterfly species
James Petiver described 48 species of British butterflies. Now 56 are known (out of 17, 500 around the world), but species are vanishing with habitat loss.

In 1720, English instrument-maker **Jonathan Sisson** (1690–1747) **added a telescopic sight to the theodolite**, paving the way for the first accurate regional surveys and maps. The first theodolite had been invented by Leonard Digges in 1554 (see 1551–54) but theodolites equipped with a telescope could be used to measure angles over long distances. It meant that the height and position of every feature in the landscape could be surveyed by the method of triangulation, which uses simple trigonometry.

English astronomer Edmond Halley makes **first practical diving bell**

1717 James Petiver publishes his catalog of British butterflies

15 May 1718 James Puckle invents the **Puckle Gun**

1720 Jonathan Sisson invents the **telescopic theodolite**